MILITARY SPECIFICATION

TEST SET, DATA LINK CONTROLLER TS-3531/ASM

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for the TS-3531/ASM Test Set, Data Link Controller, herein after referred to as the test set. The test set is for flightline use and performs operational tests of the command and control functions of Data Link Controller P/N's 475AS600 and 475AS700 available at the aircraft interface for Guided Weapons Monitor-Control Sets AN/AWW-7A and AN/AWW-7D.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of Department of Defense Index of Specification and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst,NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Federal

PPP-T-60	Tape, Packaging Waterproof
PPP-T-76	Tape Pressure Sensitive Adhesive, Paper (For Carton Sealing)
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiberboard
QQ-S-781	Strapping, Steels, and Seals

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Airborne Electronic Equipment
MIL-T-18303	Test Procedures: Preproduction and Inspection, for Aircraft Electron c Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Assoc iated Equipment
MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for
MIL-H-46855	Human Engineering Requirements for Military System Equipment and Facilities

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit loads (40″ x 48″ 4 Way Partial and 4 Way Pallets)
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-965	Parts Control Program
MIL-STD-1188	Commercial Packaging of Supplies and Equipment
MIL-STD-1472	Human Engineering Design Criteria for Military Systems Equipment and Facilities

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

PUBLICATIONS

Naval Air Systems Command

NAVAIR 16-30TS3531-1	Data Link Controller Test Set TS-3531/ASM

DRAWING

Naval Air Systems Command

DL 410AS200	Data List for Data Link Controller Test Set TS-3531/ASM

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Items description. The test set covered by this specification performs a partial checkout of aircraft stores station which have been modified to carry Monitor-Control Sets AN/AMW-7A and AN/AWW-7D (Pods). To perform a complete checkout, the weapon stations must also be tested with Aircraft Weapon Control Test Set AN/ASM-184B(V)-l. The TS-3531/ASM test set contains the power cable required to connect to the aircraft interface. The test set's operating power is supplied from the aircraft via this interface.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3 Parts and materials.

3.3.1 General requirements. The parts and materials specified in the drawings and related documents for use in construction of this test set were selected to ensure the fulfillment of all requirements of this specification. Any deviation by the contractor from the use of these specified parts and materials necessarily places both the burden of proof for the acceptability of the change, and the cost associated therewith, upon the contractor.

3.3.1.1 Nonstandard parts and material approval. Approval for the use of nonstandard parts and materials shall be obtained by the contractor as specified in MIL-STD-965.

3.3.2 Supplemental requirements. In the event that the contractor proposes to alter the test set design delineated by this specification and applicable drawings, the following supplemental requirements for the selection of parts and materials are applicable.

3.3.2.1 Basic conformance. Parts and materials selected for use in the test set shall be in accordance with MIL-T-28800, Parts shall be selected from those conforming to MIL-E-5400, unless the parts selected will not perform as required in the intended applications and environments specified herein in which case 3.3.1.1 applies.

3.3.2.2 Interchangeability. The test set shall meet the interchange-ability requirements of MIL-T-28800. The drawings listed in DL 41OAS2OO shall be the standard for measuring interchangeability of all applicable units, assemblies, subassemblies, and parts. .

3.4 Design and construction. The test set shall be manufactured in accordance with the drawings and subsidiary specifications listed in DL 41OAS2OO except insofar as errors of omission or commission on the drawings may be incompatible with the requirements of this specification. In any case it is a condition of final acceptance that the test set manufactured meet all the performance requirements of this specification, and the appropriate test documents.

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3.4.1 Dimensions. The dimensions of the test set shall be not greater than the following:

a.	Length	10.25 inches (26.0 centimeters)
b.	Width	11.25 inches (28.6 centimeters)
c.	Height	9.0 inches (22.9 centimeters)

3.4.2 Total weight. Total weight of test set shall be not greater than 25 pounds (11.4 kilograms).

3.4.3 Reliability.

3.4.3.1. Operational stability. The test set shall meet the performance requirements of this specification when operated continuously or intermittently for a period of at least 150 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.4.3.2 Operating life. The test set shall have a minimum total operating life of 1,000 hours with minimal servicing and replacement of parts.

3.4.3.3 Reliability in Mean-time-Between-Failure (MTBF). The test set shall have a minimum of 150 hours of mean (operating) time between failures when tested and accepted as specified in 4.5.3.

3.4.4 Maintainability. Maintainability Provisions shall be incorporated within the design to provide for service, access and repair of all assemblies. Normal routine testing, troubleshooting, 'alignment and calibrating of the test set shall be accomplished using standard test equipment normally available in Intermediate Maintenance Shops.

3.4.5 Transportability. The test set shall be transportable by air, rail, truck and ship when mounted in a shipping container as specified in Section 5.

3.4.6 Nameplates and identification marking. Serial number assignment, nameplate approval, and Identification marking shall be in accordance with MIL-N-18307.

3.4.6.1 Identification plates. Identification plates for the test set shall in accordance with DL 410AS200.

3.4.6.2 Decals, nameplates and markings locations. All decals, nameplates, and markings shall be affixed to the equipment in accordance with DL 41OAS2OO and all referenced marking specifications, documents, and drawings therein.

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3.4.6.3 Serial numbers. Serial numbers will be assigned by the procuring activity for the test sets under contract or purchase order. The contractor shall serialize sequentially all test sets which are delivered under the contract.

3.4.7 Color and finish. Color and finish of the test set shall be in accordance with the appropriate drawings of DL 41OAS2OO and MIL-T-28800.

3.4.8 Interference control. The generation of electromagnetic interference by the test set and the vulnerability of the test set to electromagnetic interference shall be within the following requirements:

3.4.8.1 Conducted emission 30 Hz to 20 KHz, DC power lease, CEO1. Conducted emissions from dc power leads shall be within the limits of MIL-STD-461, Test Method CEO1, over the frequency range of 30 Hz to 20 KHz.

3.4.8.2 Conducted emission, 30 Hz to 20 KHz, AC power leads, CEO1. Conducted emissions from ac power leads shall be within the limits of MIL-STD-461, Test Method CEO1, over the frequency range of 30 KHz to 20 KHz with the following exceptions: narrowband emissions measured on tthe 115 volt ac lead over the frequency range of 2 KHz to 12 KHz may exceed the limits of figure 12 of MIL-STD-461 by 11 db. Broadband emissions measured on the 115 volts ac lead may exceed the limits of figure 12 of MIL-STD-461 by 6 dB. Switch transients from the ac power leads shall be within the limits of figures 1 and 2 over the frequency range of 10 KHz to 50 KHz.

3.4.8.3 Conducted emissions, 20 kHz to 50 MHz, power leads, CE03. Conducted emissions from the control and signal leads shall be within the limits of MIL-STD-461, Test Method CE02, over the frequency range of 30 Hz to 20 KHz.

3.4.8.4 Conducted emissions, 20 KHz to 50 MHz, power leads, CE03. Conducted emissions from the ac and dc power leads shall be within the limits of MIL-STD-461, Test Method CE03, over the frequency range of 20 KHz to 50 MHz with the following exceptions: narrowband emissions measured on the +28 volt dc and power ground leads may exceed the applicable limits of MIL-STD-461 at 32 KHz by 16 dB and 13 dB, respectively. Switching transients from ac and dc power leads shall be within the limits of figures 1, 2, and 3.

3.4.8.5 Conducted emissions, 20 KHz to 50 MHz, control and signal leads, CE04. Conducted emissions from the control and signal leads shall be within the limits of MIL-STD-461 Test Method CE04, over the frequency range of 20 KHz to 50 MHz.

3.4.8.6 Radiated emissions, 30 Hz to 30 KHz, magnetic field, REO1. Radiated emissions shall be within the limits of MIL-STD-461, Test Method REO1, over the frequency range of 30 Hz to 30 KHz.

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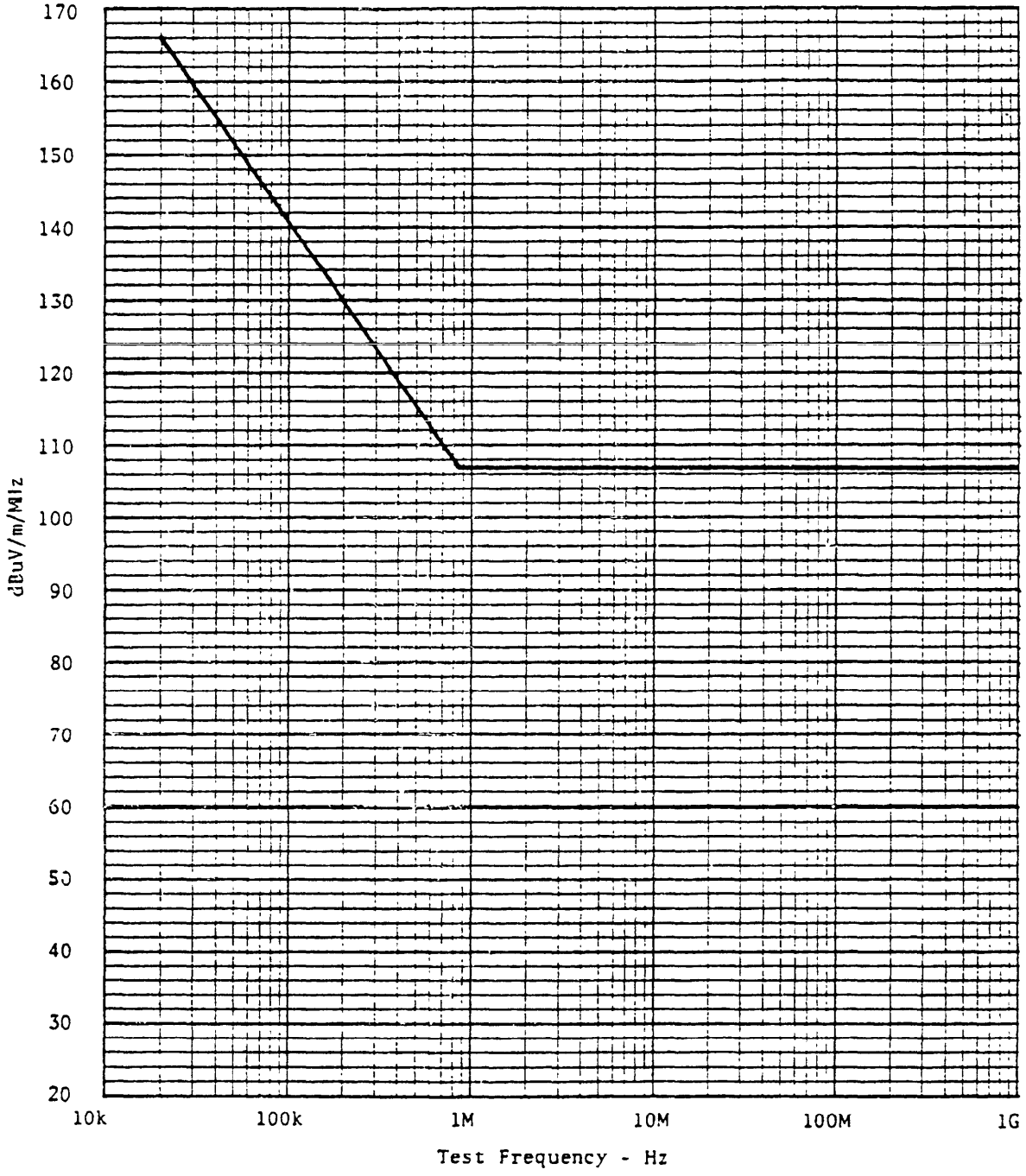


FIGURE 1. Emission limits for conducted emissions measured on 115 volt ac line during switching of test set power ON/OFF switch

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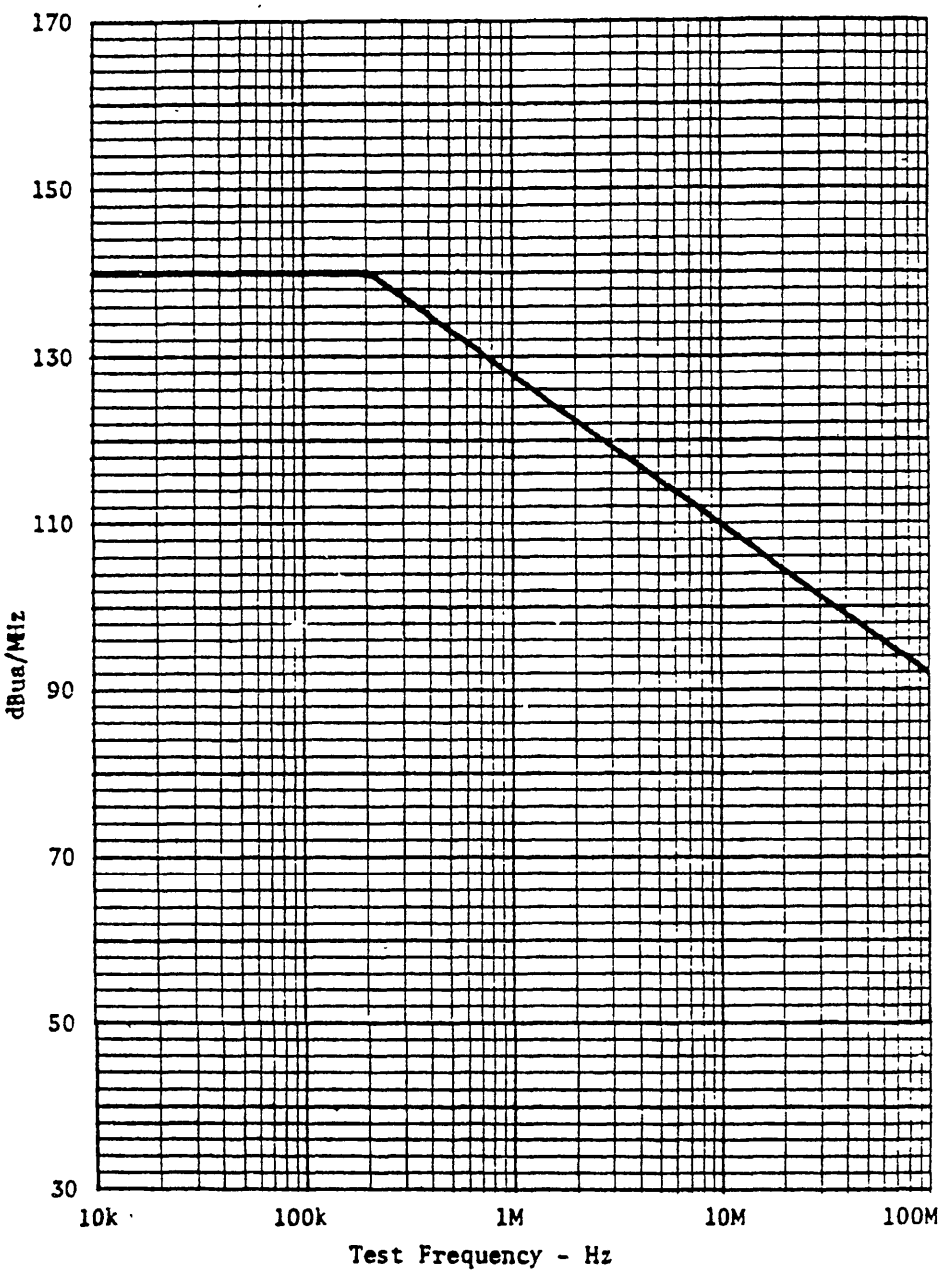


FIGURE 2. Emission limits for conducted emissions measured on power ground lines during switching of test set power ON/OFF switch

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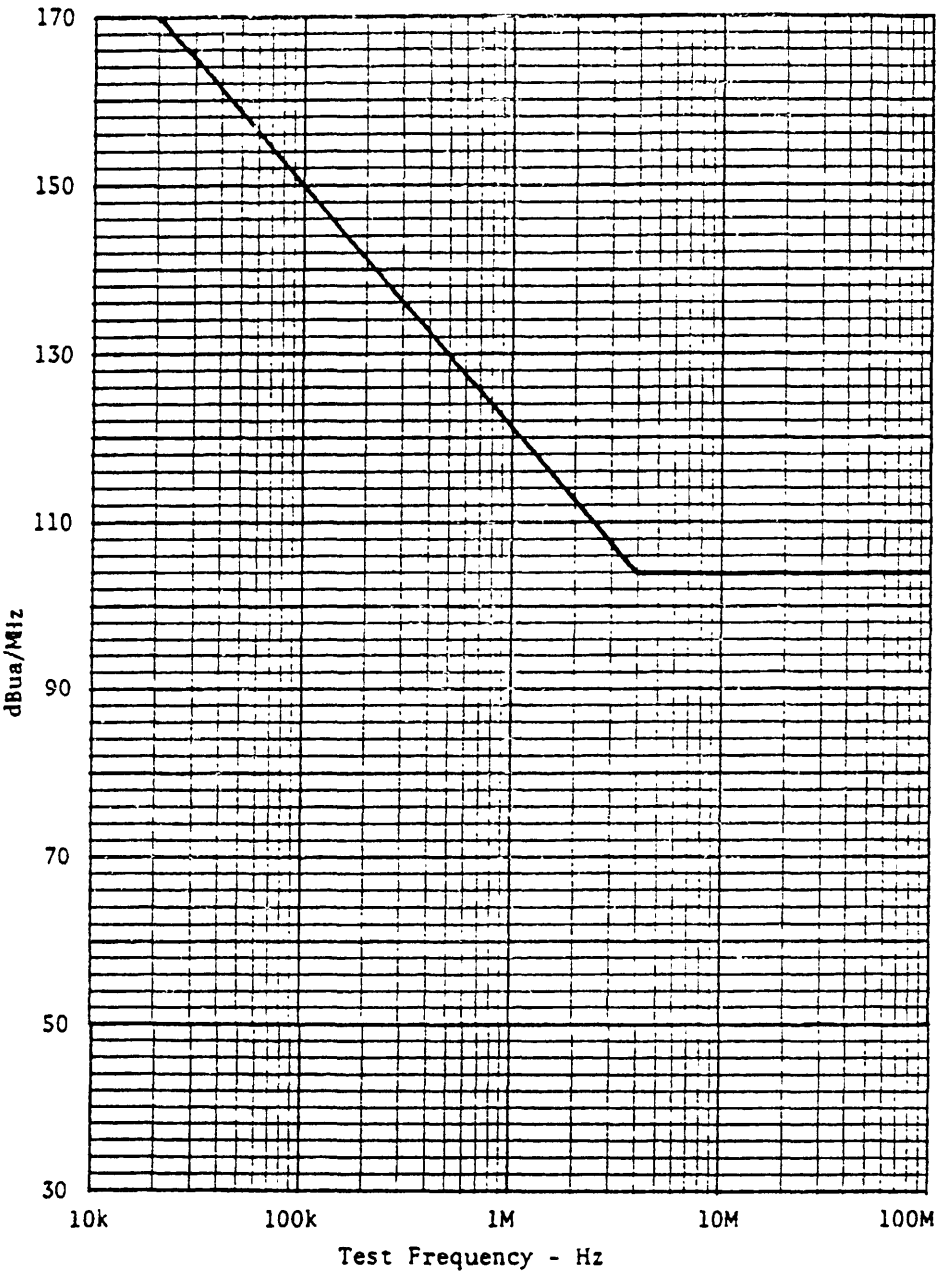


FIGURE 3. Emission limits for conducted emissions measured on 28 volt dc line during switching of test set power ON/OFF switch

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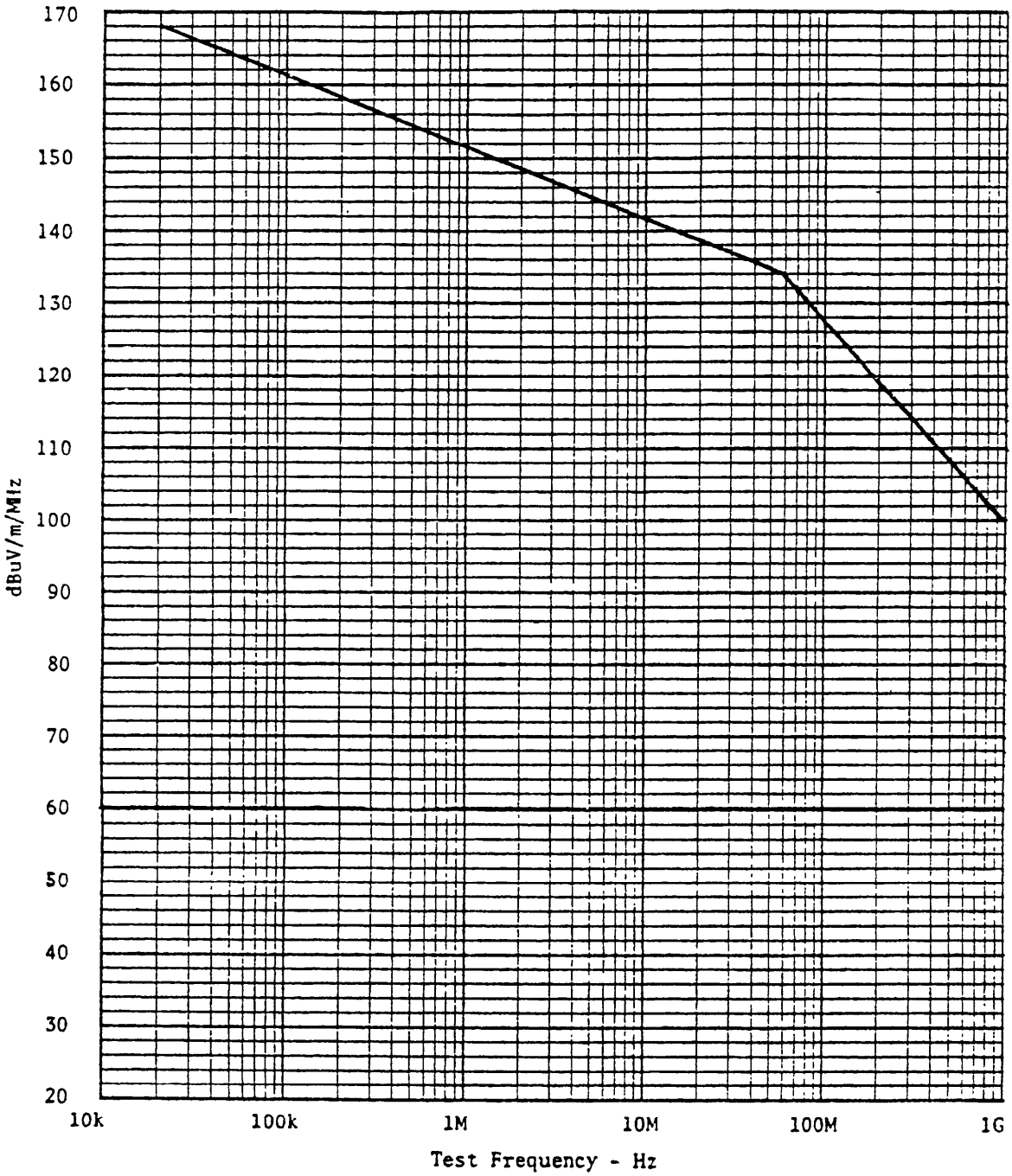


FIGURE 4. Emissions limits for radiated emissions measured during switching of test set switches

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3.4.8.7 Radiated emissions, 14 KHz to 10 GHz, electric field, RE02. Radiated emissions shall be within the limits of MIL-STD-461, Test Method RE02, over the frequency range of 14 KHz to 10 GHz. Radiated emissions due tc switching transients shall be within the limits of figure 4 over the frequency range of 14 KHz to 10 GHz.

3.4.8.8 Conducted susceptibility, 30 Hz to 50 KHz, power leads, CSO1, The test set shall not be susceptable to signal of 30 Hz to 50 KHz injected on 28 volt dc and 115 volt ac power leads at rms voltage levels of figure 5.

3.4.8.9 Conducted susceptibility, 50 KHz to 400 MHz, power leads, CS02. The test set shall not be susceptable to signals from 50 KHz to 400 MHz injected on the 28 volt dc and 115 volt ac power leads at rms voltage levels of less than 0.5 volts from a 50 ohm source.

3.4.8.10 Conducted susceptibility, spike, power leads, CS06. The test set shall not be susceptible to spike interference injected on the dc and ac power leads in accordance with MIL-STD-462, Test Method, CS06.

3.4.8.11 Radiated susceptibility, 30 Hz to 30 KHz, magnetic field, RS01. The test set shall not be suseptible to radiated signals from 30 Hz to 30 KHz as specified in MIL-STD-461 Test Method RSO1.

3.4.8.12 Radiated susceptibility, magnetic induction field, RS02. The test set shall not be susceptible to radiated fields as specified in MIL-STD-461 Test Method RS02.

3.4.8.13 Radiated susceptibility, 14 KHz to 10 GHz, electric field, RS03. The test set shall not be susceptible to radiated fields as specified in MIL-STD-461 Test Method RS03, except radiated levels shall be as indicated in table 1.

TABLE I. Radiated susceptibility levels

Frequency Range	Signal Level
10 KHz - 1.9 MHz	1 V/M
2.0 MHz - 29.99 MHz	5 V/M
30 MHz - 39.99 MHz	10 V/M
40 MHz - 97.99 MHz	4 V/M
98 MHz - 100 MHz	10 V/M
100.01 MHz - 119.99 MHz	1.2 V/M
120.01 - 140 MHz	7.5 V/M
140.01 MHz - 919.99 Mhz	0.79 V/M
920 Mhz - 12.4 GHz	5 V/M

3.4.9 Safety. The contractor shall ensure that safety requirements are in accordance with MIL-T-28800.

3.4.9.1 Human performance/Human engineering. The test set and all component parts thereof, shall be designed in accordance with the requirements of MIL-STD-1472. Recognized principles of human engineering as specified in MIL-H-46855 shall be followed during design.

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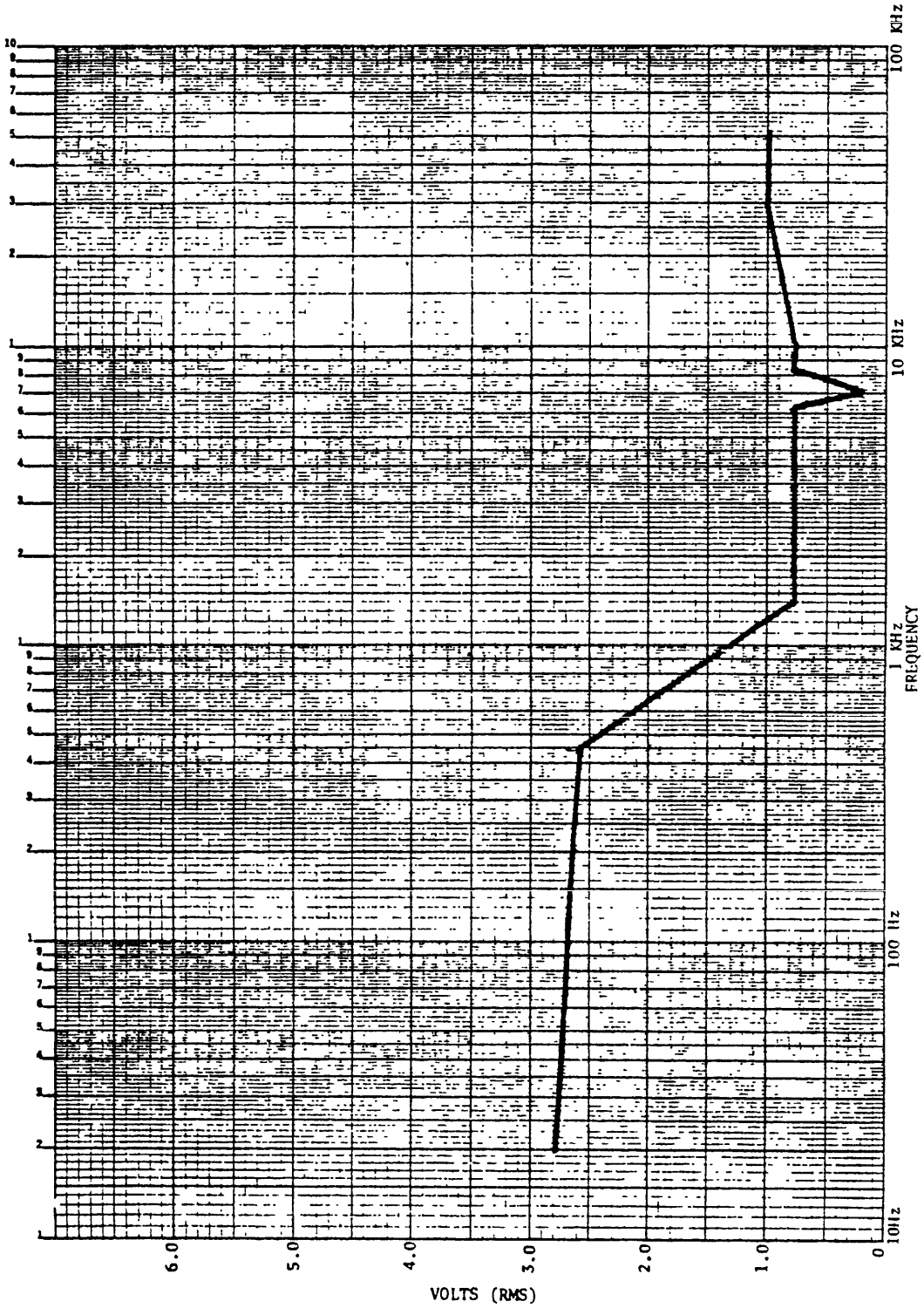


FIGURE 5. Maximum levels for conducted susceptibility test

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3.4.10 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a.	Temperature	Room ambient (30 ± 10°C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient
e.	Input power voltage	115 ± 1.5 volts, 400 ± 5 Hz ac +28 ±4 volts dc

3.4.11 Service conditions. The test equipment shall meet the performance requirements of this specication under any of the environmental service conditions or combination of these conditions as specified in MIL-T-28800 for class 2 equipment.

3.4.12 Warm up time. The time required for the equipment to warm-up prior to operations shall be not greater than 2.5 minutes under standard and extreme service conditions.

3.4.13 Primary input power. The test set shall be capable of meeting the performance requirements of this specification when supplied input power having characteristics as specified in MIL-T-28800. The power required shall be of the following types and shall be not greater than the specified amount.

a. AC Power, Single Phase, 115V, 380-420 Hz 30 VA.

b. +28 Volts DC at 1.5 ampere capacity.

3.4.13.1 Protection. The equipment shall be capable of withstanding input variations as imposed by abnormal electrical system operations as specified in MIL-STD-704.

3.5 Performance. The performance of the test set shall be as set forth below plus any and all performance requirements stated on the drawings of DL 41OAS2OO. Unless otherwise specified herein, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variations are specified herein.

3.6 Detail requirements.

3.6.1 General. Test Set, Data Link Controller TS-3531/ASM shall be manufactured in accordance with the documents listed on DL 41OAS2OO and shall meet the requirements stated thereon and in the following paragraphs.

3.6.1.1 Function. The test set shall test the functions which are peculiar to the installation of Data Link Controller PN's 475AS600 and 475AS700. All tests shall be made through the umbilical interface which

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normally mates with the data link pod. The test set shall provide a GO, NO-GO indication confirming that the-data link signals supplied to the data link pod are within the frequency ranges specified herein.

3.6.1.2 Electrical. The test set shall meet the following electrical requirements:

3.6.1.2.1 Multiplex input. The test set shall provide a virtual ground input for the current source signal drive originating in the Data Link Controller. The test set shall decode the signals when the current components are of the magnitudes specified in table II.

3.6.1.2.2 Input power. The test set shall meet performance requirements of this specification when 115 VAC, 400 Hz and +28 VDC input power from the aircraft is in accordance with MIL-STD-704.

3.6.1.2.3 Test functions. The test of all functions shall be accomplished by selecting the appropriate function and activating the PUSH/TEST switch. The specified result shall be indicated on the QUALITY meter. (References to the meter areas identified by triangles, squares or circles means the entire boxed-in area containing the symbol.)

3.6.1.2.4 Discrete functions, FUNCTIONS switch positions 1 through 5, PUSH TEST switch position A. For positions through 5 of the FUNCTION and PUSH TEST switch in position A the-QUALITY meter indicator shall move from the no signal area, defined by▽, to an active signal indicating area when the appropriate voltage controlled oscillator (VCO) in the control group is activated. The results shall be a GO indication with the VCO currents and frequencies listed in table III. A GO indication is defined as a reading with the limits of the meter area defined by l . For frequencies outside of the bands listed in the NO-GO column of table III the indicator of the QUALITY meter shall indicate a NO-GO condition (outside the ◯ area).

TABLE II. Frequencies and respective current components of signals originating in the Data Link Controller

Nominal Frequency (Hz}	Current Component (ma-peak to peak)
1,300	0.28 ± .056
1,700	0.28 ± .056
5,400	0.33 ± .066
7,350	0.50 ± .100
10,500	0.77 ± .150
22,000	2.0 ± .400
30,000	3.3 ± .660

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TABLE III. Function switch positions and respective GO/NO-GO frequency tolerances and nominal currents

Function Switch Position	GO Frequency (Hz)	Nominal Current Ma P-P (Ref)	NO-GO Frequency (Hz)
1	1,300 ±26	.28	1,248 < f < 1,352
2	1,700 ±34	.28	1,632 < f < 1,768
3	5,400 ±108	.33	5,184 < f < 5,616
4	7,350 ±148	.50	7,054 < f < 7,646
5	10,500 ±210	.77	10,080 < f < 10,920

3.6.1.2.5 Proportional functions, FUNCTION switch positions 6 and 7, PUSH TEST switch position A. he test set shall decode two proportional deviation multiplex channels and provide a GO, NO-GO indication of deviation and polarity. The frequencies associated with the proportional channels and the frequencies and current magnitudes required to give a GO indication are shown in table IV. The test set shall give a GO indication throughout the tolerance range in the FREQUENCY column.

3.6.1.2.6 Internal power supply self-test, FUNCTION switch positions 8 through 11, PUSH TEST switch position A. he test set shall provide a GO, NO-GO indication of the internal (test set) power supply voltages in accordance with the data provided in table V.

3.6.1.2.7 Channel select functions, FUNCTION switch positions 1 through 4, PUSH TEST switch position B. he two channel select inputs (pins J1-25 and J1-26) are either +28 VDC or 0 VDC. The test set shall decode any combination of these inputs and give indications as shown in table VI.

3.6.1.2.8 Pod status indication (PSI) function, FUNCTION switch position 5, PUSH TEST switch position B. The PSI input (J1-11) is either +28 VDC or 0 VDC. The test set shall indicate in the △ area for +28 VDC at J1-11 and indicate in the ● area for 0 VDC at J1-11.

3.6.1.2.9 Audio function, FUNCTION switch position 6, PUSH TEST switch position B. The test set shall detect the presence of a 2V p-p audio input and give a meter indication to the right.

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TABLE IV. Frequencies and current magnitudes required for a GO indication

Function Switch Position	Frequency (Hz)	Nominal Current Ma P-P (Ref)	GO, Meter Indication
6	22,000 ±150	2.0	●
6	23,650 ±330	2.0	▭
6	20,350 ±330	2.0	▬
7	30,000 ±200	3.3	●
7	32,250 ±450	3.3	▭
7	27,750 ±450	3.3	▬

TABLE V. GO/NO-GO indications for test set interval power supply voltages

Function Switch Position	Voltage Volts DC	GO Meter Indication
8	+5.0 ±0.3	▽
9	+12.0 ±0.5	▽
10	-5.0 ±0.3	▼
11	-12.0 ±0.5	▼

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TABLE VI. Test set indications for decoded combination of inputs

J1-25	J1-26	Meter Indication for Function Switch Position			
		1	2	3	4
0 VDC	0 VDC	☐	●	●	●
28 VDC	0 VDC	●	☐	●	●
0 VDC	28 VDC	●	●	☐	●
28 VDC	28 VDC	●	●	●	☐

3.6.2 Content. The TS-3531/ASM Test Set, DL 410AS200, shall consist of the following:

3.6.2.1 Panel assembly. This assembly shall contain all of the electronics and controls for performing the test functions. The panel assembly shall be manufactured in accordance with DL 410AS200.

3.6.2.2 Power cable assembly. The power cable assembly shall be manufactured in accordance with DL 410AS200. This cable shall be designed to connect the test set to the aircraft being used.

3.6.2.3 Service cable assembly. The service cable assembly shall be manufactured in accordance with DL 410AS200. This cable shall be used for test set maintenance, calibration, alignment and fault isolation. The cable shall be designed to provide terminals for bench test inputs and power application.

3.6.2.4 Spanner wrenches. The spanner wrenches shall be manufactured in accordance with DL 410AS200. Two spanner wrenches shall be furnished with the test set. The spanner wrenches shall be designed for disassembly and assembly of power cable assembly.

3.6.2.5 Case and cover. The case and cover shall be manufactured in accordance with DL 410AS200. The cover shall be designed for storage of switch box and service cable, power cable and two spanner wrenches.

3.7 Workmanship. Workmanship shall be in accordance with the requirements of DL 410AS200 and MIL-T-28800.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4) .

b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 3.4.10.

4.4 First article inspection. First article inspection shall be made on a test set representative of the production equipments to be supplied under the contract. First article inspection shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedures (see 4.11). Column I of table VII lists minimum tests required for first article inspection. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government. No first article inspections shall be conducted prior to acceptance of first article test procedure by the procuring activity (see 6.2.2).

4.4.1 First article test data. The contractor shall make available all data collected in conducting these tests for procuring activity review and acceptance (see 6.2.2). This data including a list of all electrical and electronic parts giving their specified voltage, and ambient and surface temperatures shall be made available on first production contracts. The ambient and surface temperatures shall be obtained under the extreme high temperature operating condition.

4.4.2 Scope of first article tests. First article tests shall include all tests listed in table VII, column I and any additional tests deemed necessary by the procuring activity to determine that the test set meets all the requirements of this specification.

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TABLE VII. Test requirements

Requirement	I. First article Inspection	II. Quality Conformance Inspections			
		Individual (A)	Sampling (B)	Special (C) 1/	Life (D)
Visual Examination (3.3, 3.4, 3.4.1)	4.5.1.1	4.5.1.1	4.5.1.1	4.5.1.1	4.5.1.1
Performance Characteristics (3.5)	4.5.1.2	4.5.1.2	4.5.1.2	4.5.1.2	4.5.1.2
Workmanship (3.7)	4.5.1.3	4.5.1.3	4.5.1.3	4.5.1.3	4.5.1.3
Environmental (3.4.11)					(See 4.6.1, 4.6.2 and 4.6.3)
Warm-up time (3.4.12)		1/	2/	3/	
Power protection (3.4.13.1)					
Interference Control (3.4.8)	4.7				
Reliability (3.4.3)	4.5.3		4.5.3.2		

1/ Additional individual tests may be requested by the procuring activity (see 4.5.1 and 6.2.1).

2/ Sampling tests to be performed will be determined by the procuring activity based on first article and individual test data (see 4.5.2.1).

3/ Special tests to be performed will be determined by the procuring activity (see 4.5.4.2).

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4.4.3 First article approval. Approval of first article sample shall be by the procuring activity upon satisfactory completion of all tests listed in table VI column I and the first article test report being made available (see 6.2.2). No production test sets shall be delivered prior to the approval of the first article sample.

4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance inspection (see 6.2.2). All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall prepare test reports showing quantitative results for all quality conformance tests (see 6.2.2). Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance inspection shall consist of the following tests:

a. Individual Tests (see 4.5.1)

b. Sampling Tests (see 4.5.2)

c. Reliability Assurance Tests (see 4.5.3)

d. Special Tests (see 4.5.4)

e. Life Tests (see 4.6)

4.5.1 Individual tests. Each test set submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each test set accepted shall have passed the following tests:

a. Examination of Products

b. Operational Test

c. Manufacturing Run-in Test

4.5.1.1 Examination of product. Each test set shall be examined carefully to determine that the material and workmanship requirements have been met.

4.5.1.2 Operational test. Each test set shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure the equipment operation meets the requirements of this specification.

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4.5.1.3 Manufacturing run-in test. Each test set shall be operated under the conditions specified herein for a period of six hours without failure. A failure shall be defined as anything which causes malfunctioning of the test set. Only those adjustments will be permitted which can be made by using controls and adjustments that are accessible to the operator during the normal use of the test set.

a.	Temperature	Ambient room
b.	Humidity	Ambient room
c.	Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3g's

The test set shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the six-hour period of operation. Where feasible, the test set shall be operated during this vibration for the purpose of detecting flaws and imperfect workmanship. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to that plane for five minutes. Where it is not feasible to vibrate the test set in two directions the vertical direction shall be used. During the six-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future test sets. A record shall be kept of all failures. The 6-hour period specified above may be composed of two three-hour periods to conform with standard working hours.

4.5.2 Sampling tests. Test sets selected for sampling tests shall first have passed the individual tests. Test sets shall be selected for sampling tests by the government inspector in accordance with table VIII.

4.5.2.1 Scope of sampling tests. As a minimum, each test set selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each equipment tested and shall be based on results of the first article, individual, and special tests.

c. Manufacturing run-in test specified in 4.5.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future test sets.

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4.5.3 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with MIL-STD-781. Tests as required by both the qualification phase and the production acceptance (sampling) phase shall be conducted.

TABLE VIII. Sampling tests sample selection

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

4.5.3.1 Qualification phase. Prior to the acceptance of test sets under the contract or purchase order. a minimum of three test sets shall be tested in accordance with MIL-STD-781; under the section entitled “Qualification Phase of Production Reliability Tests”. The maximum number of test sets to be used shall be those listed in table V of MIL-STD-781. For the qualification phase, Test Level E shall be used. The accept-reject criteria for Test Plan IV shall be used.

4.5.3.2 Reliability sampling phase tests. Samples of the test set shall be tested in accordance with MIL-STD-781 under the section entitled “Sampling Phase of Production Reliability Tests”. For the sampling phase, Test Level E shall be used, except that the test set shall operate continuously. The accept-reject criteria for Test Plan V shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.5.3.2.1 Lot size for sampling phase. The test sets constructed during a two month period shall be defined as one lot. The accept-reject criteria for Test Plan IV shall be used to determine the length of the tests in each lot. Unless otherwise specified in the contract or purchase order, no test sets shall be shipped until an accept decision is reached under each lot (see 6.2.1).

4.5.3.3 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, and preventive maintenance to be allowed during the test, shall be part of the test procedures to be made available and approved by the procuring activity prior to the beginning of the qualification test phase of the reliability assurance tests (see 6.2.2).

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4.5.4 Special tests. Special tests shall be conducted on a quantity of test sets for the purpose of checking the effect of any design or material change on the performance of the test set and to assure quality control. The test set selected for special tests may be selected from test sets previously subjected to the sampling or reliability assurance tests.

4.5.4.1 Special tests schedule. Selection of test sets for special tests shall be made as follows:

a. On an early test sets, after an engineering or material change.

b. Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.5.4.2 Scope of special tests. Special tests shall consist of such tests as are acceptable to the procuring activity. Test procedures previously approved for the first article tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure for procuring activity acceptance prior to conducting the tests (see 6.2.2).

4.5.5 Equipment failure. Should a failure occur during either the sampling or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

c. Have available, for procuring activity acceptance, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests (see 6.2.2).

d. Where practical, include a test in the individual test to check all test sets for this requirement until assurance is obtained that the defect has been corrected.

4.6 Life test. The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on test sets that have passed the individual tests. The life test shall be performed under the conditions specified in 4.6.1. The life test sample shall be selected by the government inspector in accordance with the following. (Test sets which have successfully passed the Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of test sets equal to, or more than, that listed in table IX receive at least 300 hours each of test time.

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TABLE IX. Life test sample selection

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.6.1 Test conditions. The life test shall be conducted under the standard conditions specified in 3.4.10.

4.6.2 Test periods. The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the test set shall be turned on and off several times and put through 'its various phases of operation.

4.6.3 Performance check. At approximately eight-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to authorization by the procuring activity (see 6.2.2).

4.6.4 Test data. The contractor shall keep a daily record of the performance of the test set, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test (see 6.2.2). This record shall indicate the following:

a. Part type number

b. The circuit reference symbol number

c. The part function

d. Name of the manufacturer

e. Nature of the failure

f. The number of hours which the part operated prior to failure

4.6.4.1 Failure report. In the event of a failure, the government inspector shall be notified immediately. A report shall be prepared and made available to the procuring activity upon completion of the test (see 6.2.2). In this report, the contractor shall propose suitable and adequate design or

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material corrections for all failures which occurred (see 6.2.2), The procuring activity will review such proposals and determine whether they are acceptable.

4.7 Electromagnetic interference (EMI) tests. EMI tests shall be performed in accordance with MIL-STD-462 to determine that the test set meets the requirements of MIL-STD-461 for Class Al equipment (see 3.4.8).

4.8 Environmental tests. The test set shall be subjected to environmental tests specified in MIL-T-28800 for Class 2 equipment.

4.9 Warm-up time. The test set shall be subjected to tests to determine the capability of the test set to conform to the warm-up time requirements of 3.4.12.

4.10 Input power test.

4.10.1 Voltage and frequency variation tests. The test set response to the input voltage and frequency variations shall be tested in accordance with MIL-T-28800. Operational performance may be degraded during the exposure but test set shall not sustain damage and shall return to normal operation when test is completed.

4.10.2 Voltage and frequency-transient test. The test set performance during voltage transients shall be tested in accordance with MIL-T-28800. Operational performance may be degraded during the exposure but test set shall not sustain damage and shall return to normal operation when test is completed.

4.11 Test procedures. The procedures used for conducting first article tests, quality conformance tests, and life tests shall be prepared by the contractor for procuring activity review and acceptance (see 6.2.2). The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures shall be provided and may be used when their use is authorized by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests when deemed necessary (see 6.2.1).

4.12 Presubmission testing. No item, part or complete test set shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements.

4.13 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of Section 5 and the standards and specifications listed therein.

5. PACKAGING.

5.1 Preservation. Preservation shall be level A or industrial as specified (see 6.2.1).

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5.1.1 Level A.

5.1.1.1 Cleaning. Test sets shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Test sets shall be dried in accordance with MIL-P-116.

5.1.1.3 Unit packs. Unless otherwise specified in the contract, (see 6.2.1), each test set shall be individually unit packed in accordance with the methods of MIL-P-116 designated herein ensuring compliance with the applicable requirements of that specification.

5.1.2 Industrial. The industrial preservation of test sets shall be in accordance with the requirements of MIL-STD-1188.

5.2 Packing. Packing shall be level A or industrial, as specified in the contract (see 6.2.1).

5.2.1 Level A. The packaged test sets shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the Appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joints with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the Appendix to PPP-B-636 using nonmetallic or tape banding only. For the Army, containers shall conform to PPP-B-601, overseas type; PPP-B-621, Class 2, Style 4 or PPP-B-585, Class 3, Style 2 or 3. Closure and strapping shall be in accordance with applicable container specifications except the strapping shall conform to QQ-S-781, Type I or IV, Finish A. When the gross weight exceeds 200 pounds or the container length and width is 48 inches x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification.

5.2.2 Industrial. The preserved test sets shall be packed in accordance with the requirements of MIL-STD-1188.

5.2.3 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or purchase order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.3.1 Level A. Test sets, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.3 Marking.

5.3.1 Level A. In addition to any special or other identification marking specified in the contract (see 6.2.1), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

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5.3.2 Industrial. Industrial marking shall be in accordance with the requirements of MIL-STD-1188.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1 and 5.2.2) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.15.

6. NOTES

6.1 Intended use. The TS-3531/ASM Data Link Controller Test Set is a small portable flight-line test device used to perform preflight and periodic performance evaluation of Data Link Controller P/N's 475AS600 and 475AS700. The controller initiates certain data link commands, and proportional signals for Guided Weapon Monitor-Control Sets AN/AWW-7A and AN/AWW-7D. The test set is used in conjunction with Aircraft Control Test Set AN/ASM-184B(V)-1 to provide an evaluation of the Aircraft interface for the Guided Weapons Monitor-Control Sets AN/AWW-7A and AN/AWW-7D. The control group output frequencies can be evaluated with the test set with the test result displayed on a front panel mounted quality meter.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of packaging and packing (see 5.1).

co When a first article is required for inspection and approval (see 4.4).

d. Additional individual tests, if required (see 4.5.1 and table VIII).

e. Lot acceptance for each lot, if other than specified herein (see 4.5.3.2.1).

f. Test procedure modifications or additional tests, if required . (see 4.11).

g. Unit packs, if other than specified herein (see 5.1.1.3).

6.2.2 Data requirements. When this specification is used in an acquisition which Incorporates a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in

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accordance with the approved CDR incorporated into the contract. When the provisions of DAR 7-104.9 (N) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.
4.4, 4.11	Procedure, First Article Test	DI-T-4901
4.4.1, 4.4.3	Report, First Article Test	DI-T-4902
4.5, 4.11	Procedure, Quality Conformance Test	DI-T-4903
4.5	Report, Quality Conformance Test	DI-T-4904
4.5.3.3	Procedure, Reliability Assurance Test	DI-R-7035
4.5.4.2	Procedure, Special Test	DI-T-4903
4.6.4.1	Engineering Change Proposal	DI-E-2037
4.5.5c	Proposal, Corrective Action	UDI-T-23719
4.6.3, 4.11	Proposal, Life Test	DI-T-4903
4.6.4	Report, Life Test	DI-T-4904
4.6.4.1	Report, Failed item analysis	DI-R-7039

(Data item descriptions related to this specification, and identified in section 6 will be-approved and listed as such in DOD 5000.19L. , Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval of the documents' first article.

Preparing Activity
NAVY-AS
(Project 4935-N047)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-85378(AS)	TEST SET, DATA LINK CONTROLLER TS-3531/ASM, Specification

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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